Inequalities WS 2

1. If $a \geq b$, prove that $a^3 - b^3 \geq a^2b - ab^2$.
2. It is known that $a \leq |a|$ for any real number a. Let x and y be any two real numbers. Use the above result to prove the triangle inequality $|x+y| \leq |x| + |y|$. Begin by squaring both sides.
3. For any real numbers $\mathrm{x, y, z, u}$ prove that: $(\mathrm{x}^2 + \mathrm{y}^2)(\mathrm{z}^2 + \mathrm{u}^2) \geq (\mathrm{xz} + \mathrm{yu})^2$
4. Suppose that $m \leq f(x) \leq M$ in the interval $a \leq x \leq b$. Use a diagram to help prove that $m(b-a) \leq \int_a^b f(x)\,dx \leq M(b-a)$.
5. If $a > 0, b > 0$ and $c > 0$,
    a. Show that $(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) \geq 9$.
    b. Deduce that $(a+b+c)(ab+bc+ca) \geq 9abc$
    c. Hence deduce that $a^2b + b^2c + c^2a + ab^2 + bc^2 + ca^2 \geq 6abc$.
6. If $a > 0, b > 0, c > 0$ and $d > 0$,
    a. Show that $(a+b+c+d)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right) \geq 16$
    b. Hence, show that $\frac{16}{a+b+c+d} \leq \frac{3}{b+c+d} + \frac{3}{a+b+d} + \frac{3}{a+b+c} + \frac{3}{a+c+d} \leq \frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}$
7. x, y, z are positive real numbers
    a. Prove that $x^2 + y^2 + z^2 \geq xy + yz + xz$
    b. Given that the inequality $\frac{1}{x}+\frac{1}{y}+\frac{1}{z} \geq \frac{9}{x+y+z}$ holds and $x^2+y^2+z^2 = 9$, prove that $\frac{1}{1+xy} + \frac{1}{1+yz} + \frac{1}{1+xz} \geq \frac{3}{4}$
8. The curve $y = f(x)$ is concave up for all real values of x. a, b, c and d are the x-coordinates of four points lying on this curve
    a. Show that $f(a) + f(b) > 2f\left(\frac{a+b}{2}\right)$
    b. Show that $f(a) + f(b) + f(c) + f(d) > 4f\left(\frac{a+b+c+d}{4}\right)$
    c. Consider the curve $y = (x+1)^{-2p}$, where p is a positive integer and by using the above, or otherwise, show that $\frac{1}{(n+2)^{2p}} + \frac{1}{(n+4)^{2p}} + \frac{1}{(n+6)^{2p}} + \frac{1}{(n+8)^{2p}} > \frac{4}{(n+5)^{2p}}$
9. a, b, c, d are positive real numbers, each strictly less than $\frac{\pi}{2}$
    a. Show that $\cos a \cos b \leq \left[\cos\frac{1}{2}(a+b)\right]^2$
    b. Hence show that $\cos a \cos b \cos c \cos d \leq \left[\cos\frac{1}{4}(a+b+c+d)\right]^4$
    c. Deduce that $\cos a \cos b \cos c \leq \left[\cos\frac{1}{3}(a+b+c)\right]^3$
10. a. If $\mathrm{f(x), g(x)}$ and $\mathrm{h(x)}$ are distinct non-negative continuous functions of x in the interval $\mathrm{a \leq x \leq b}$ and $\mathrm{f(x) < g(x) < h(x)}$, explain why $\int_a^b \mathrm{f(x)dx} < \int_a^b \mathrm{g(x)dx} < \int_a^b \mathrm{h(x)dx}$
    b. By considering the interval $0 < \mathrm{x} < 1$ as an inequality, use algebra to show that $\frac{1}{2}\mathrm{x}(1-\mathrm{x})^3 < \frac{\mathrm{x}(1-\mathrm{x})^3}{1+\mathrm{x}} < \mathrm{x}(1-\mathrm{x})^3$
    c. Deduce that $\frac{1}{2}\int_0^1 \mathrm{x}(1-\mathrm{x})^3\mathrm{dx} < \int_0^1 \frac{\mathrm{x}(1-\mathrm{x})^3}{1+\mathrm{x}}\mathrm{dx} < \int_0^1 \mathrm{x}(1-\mathrm{x})^3\mathrm{dx}$
    d. Show that $\int_0^1 \frac{\mathrm{x}(1-\mathrm{x})^3}{1+\mathrm{x}}\mathrm{dx} = \frac{67}{12} - 8\ln 2$, hence deduce that $\frac{83}{120} < \ln 2 < \frac{667}{960}$
11. a. Show that $\binom{\mathrm{n}}{\mathrm{r}} < \mathrm{n}\binom{\mathrm{n}-1}{\mathrm{r}-1}$, where $\mathrm{n, r} \in \mathbb{Z}^+$ and $1 < \mathrm{r} \leq \mathrm{n}$.
    b. Given that $(\mathrm{a}+\mathrm{b})^\mathrm{n}$ can be written as $\binom{\mathrm{n}}{0}\mathrm{a}^\mathrm{n} + \binom{\mathrm{n}}{1}\mathrm{a}^{\mathrm{n}-1}\mathrm{b} + \cdots + \binom{\mathrm{n}}{\mathrm{n}}\mathrm{b}^\mathrm{n}$, show that for $\mathrm{a, b} > 0$ and $\mathrm{n} \in \mathbb{Z}^+$, $(\mathrm{a}+\mathrm{b})^\mathrm{n} - \mathrm{a}^\mathrm{n} < \mathrm{nb}(\mathrm{a}+\mathrm{b})^{\mathrm{n}-1}$